



SCHOOL OF INFORMATION TECHNOLOGY

Assignment Title	SF1
Assignment Type	Project
	Individual Assignment
Total Marks	100

Module Name	Web Animation Scripting
Module Code	WAS262

- Consult the Learning Management System (LMS), i.e., Canvas or ECI, for the due dates, the Assessment Guidelines for Students document, the Assignment Overview, and other updated detail. Check the Announcements by your lecturer and use the Module Q&A forum to ask questions related to this assignment.
- For assessment rules and regulations, refer to the Student Handbook.
- Note STADIO's rules relating to plagiarism and cheating. Do not hesitate to seek help if you are unsure of how to reference your sources. Make use of the STADIO Referencing Guide, available from the Online Library.

INTRODUCTION

Welcome to the WAS262 module, the goal of this module is to equip you with the essential skills and knowledge to create and build animated elements for your web projects and applications. Using HTML5, CSS, and JavaScript in Canvas, you will learn how to develop engaging and dynamic animations. These techniques are widely used in the industry and mastering them will enhance your ability to deliver professional and captivating web experiences. Let's dive into the world of web animations and bring your projects to life!

OVERVIEW

To successfully complete this assessment, you must follow the specific criteria and requirements aligned with Learning Objectives 1 to 4. Your task is to create a JavaScript-based animation using HTML5 Canvas, where multiple balls bounce dynamically across the screen. The animation should incorporate randomized motion using JavaScript's Math functions, include particle effects, display an animation counter, and demonstrate an understanding of selected principles from the 12 principles of animation.

Task	Description	Submission format	Marks
1	Practical	Designing & developing an interactive web-based game	100

ARTIFICIAL INTELLIGENCE TRAFFIC LIGHT MATRIX (AITLM)				
	RED	AMBER	GREEN	
Question/ Part	May NOT use AI	May use AI	MUST use AI	Detail of AI use
Task 1		●		May use for assist in project work, feedback, and editing. Where you use AI, submit evidence of all your prompts and AI outputs.
NOTE: - STADIO employs AI writing detection tools to monitor AI use. - Presenting work produced by AI as your own contravenes the STADIO Plagiarism policy. Refer to the STADIO Referencing Guide for guidance.				

ASSIGNMENT DESCRIPTION	
NO	TASKS TO BE COMPLETED
1	Task 1: PLANNING & ETHICS (20 MARKS)  Image source: (OpenAI, 2026) Context: EcoDash: African Digital Logistics & Infrastructure Simulator

Assessment Introduction

Across Africa, delivering essential goods and services to rural and underserved communities presents significant challenges. Poor road infrastructure, long travel distances, load-shedding, unpredictable weather, and limited connectivity often affect the movement of medical supplies, food, educational resources, and other essential items. At the same time, sustainable technologies such as electric vehicles, drones, and solar-powered infrastructure are becoming increasingly important in addressing these challenges.

In this assessment, you will apply your knowledge of **HTML5 Canvas**, **CSS3**, and **JavaScript (ES6+)** to design and develop **EcoDash**, an interactive 2D simulation that demonstrates how technology can improve logistics and infrastructure within an African context. Rather than creating a traditional game, you will build a simulation that models real-world scenarios through animation, user interaction, and problem-solving.

The project will require you to implement dynamic graphics, object movement, collision detection, animations, user input, mathematical calculations, and interactive gameplay mechanics using the HTML5 Canvas API. You will also follow modern software development practices by using **GitHub** for version control, documenting your work, and demonstrating your understanding of the code during project presentations. Throughout the assessment, you are encouraged to think critically about how digital technologies can solve authentic African challenges while producing a professional, well-structured, and engaging web application.

Assessment Overview

Your task is to design and develop **EcoDash**, an interactive HTML5 Canvas web application that simulates a solar-powered drone or electric delivery vehicle transporting essential supplies across an African environment. The simulation should present users with realistic logistics challenges that require planning, navigation, and resource management. As the player progresses, they must successfully complete delivery missions while responding to changing environmental and infrastructure conditions.

Examples of simulation features may include:

- Delivering medical supplies, food, educational resources, or emergency equipment.
- Navigating roads, rivers, mountains, rural villages, and urban environments.
- Avoiding obstacles such as wildlife, damaged roads, flooding, construction zones, or traffic.
- Managing battery levels and recharging at renewable energy stations.
- Responding to weather conditions such as rain, wind, dust storms, or reduced visibility.
- Adapting to load-shedding schedules that temporarily disable charging stations or infrastructure.
- Completing deliveries within specified time limits while maintaining vehicle efficiency.

The application must be developed using **HTML5 Canvas**, **Vanilla JavaScript (ES6+)**, and **CSS3** without the use of external game engines or frameworks. Students are expected to apply object-oriented programming principles, modular JavaScript, mathematical calculations for movement and collision detection, animation techniques, and responsive interface design.

In addition to technical implementation, the project should demonstrate an understanding of **African digital innovation** by incorporating authentic local challenges, environments, or sustainable technology solutions into the overall simulation.

The assessment consists of three major tasks supported by milestone monitoring sessions:

- **Task 1** (20 Marks): Project planning, GitHub repository setup, and African context research.
- **Task 2** (50 Marks): Development of the interactive simulation, including gameplay mechanics, animation, mathematics, and Africanisation features.
- **Task 3** (20 Marks): Project presentation, code defence, and a live AI-Buster challenge to demonstrate individual understanding of the implementation.
- **Monitoring** (10 Marks): Progress reviews conducted during scheduled monitoring sessions.

By completing this project, you will demonstrate your ability to develop an interactive web application that combines technical programming skills, software engineering practices, creative problem-solving, and socially relevant digital innovation.



Image Source : <https://www.corporatevision-news.com> (CorporateVision, 2022)

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TASK 1: PLANNING, GITHUB & AFRICANISATION (20 MARKS)

Before developing your application, you must plan your project, investigate an African logistics challenge, and prepare your development environment. This task focuses on project planning, research, GitHub version control, ethical AI usage, and designing the overall structure of your simulation. The evidence produced in this task will guide the development of your application in Task 2

1.1 GITHUB REPOSITORY SETUP (5 MARKS)

What you need to do:

Create a public GitHub repository for your project and use it throughout the development process. Your repository must demonstrate continuous progress through regular commits and include the required project documentation.

Create a public GitHub repository named:

EcoDash-African-Logistics

Your repository must include:

- README.md
- Project title
- Project description
- Installation/setup instructions
- AI Usage Disclosure Table
- Initial project folder structure

Important

Your repository must demonstrate continuous development.

You are required to make at least eight (8) meaningful commits on different days throughout the project. Uploading the entire project in one commit or creating the repository immediately before submission will result in 0/5 for this section.

1.2 AFRICAN PROBLEM INVESTIGATION (5 MARKS)

What you need to do

Research and describe a real-world African logistics or infrastructure challenge that your simulation will solve. Explain how mathematical concepts such as movement, vectors, or physics will be used to simulate this scenario.

Create a one-page report (Markdown or PDF) explaining:

- **Problem Context:** Identify a specific real-world African challenge your game models (e.g., *Navigating load-shedding blackouts in Johannesburg*, *Solar drone cargo deliveries in rural Rwanda*, or *Avoiding pothole networks in congested urban hubs*).
- **Physics Mapping:** Explain how trigonometry and vector math will simulate environmental challenges (e.g., crosswinds, battery drain curves, or variable terrain drag).

African Challenge

Choose one real-world problem your simulation addresses.

Examples include:

- Solar-powered medical deliveries in rural Kenya
- Load-shedding affecting deliveries in South Africa
- Flooded roads in Mozambique
- Poor road infrastructure in rural Zambia
- Wildlife crossings affecting transport routes
- Limited cellular coverage in remote communities

Mathematical Model

Explain how mathematics will be used in your simulation.

For example:

- Vehicle speed
- Direction
- Acceleration
- Battery consumption
- Wind resistance
- Terrain friction
- Collision calculations

1.3 WIREFRAME & AI REFLECTION (10 MARKS)

What you need to do

Design a wireframe showing the layout of your application and document how AI was used ethically during development. You must explain how you evaluated and improved any AI-generated suggestions instead of copying them directly.

Submit:

Wireframe

Create a sketch (hand-drawn or digital) showing:

- Main game screen
- Player
- HUD
- Score
- Battery indicator
- Buttons
- Mini-map (optional)

AI Reflection Log

Document how AI was used during development.

Include:

- The prompt you used.
- The AI response.
- Problems you identified.
- How you improved or modified the code.

Simply copying AI-generated code without understanding it will result in mark deductions.

Referencing included and correct:

- **List of References**
 - Included
 - Correct format used
- **In-Text Referencing**
 - Included
 - Correct format used

Include at least two Academic references from **Stadio Libraries**

TASK 2: GAME DEVELOPMENT (50 MARKS)

Brief Explanation

In this task, you will develop the complete EcoDash simulation using HTML5 Canvas, CSS3, and JavaScript. Your application should demonstrate animation, mathematical calculations, object-oriented programming, collision detection, user interaction, and an authentic African-inspired theme. All features should work together to create a realistic and engaging logistics simulation.

Develop the complete simulation using only:

- HTML5 Canvas
- CSS3
- Vanilla JavaScript

2.1 PLAYER MOVEMENT & PHYSICS (15 MARKS)

What you need to do

Develop the player-controlled drone or electric vehicle and implement realistic movement using JavaScript mathematics...etc. The player should move smoothly while managing energy or battery resources throughout the simulation. E.g., make use of:

- **Vector & Trig Dynamics:** Implement vertical/horizontal vehicle movement using directional velocity, thrust, drag, and angle calculations ($\cos$, $\sin$).
- **Resource Management Mechanics:** Implement a dynamic resource constraint reflective of African infrastructure (e.g., a Solar Reserve / Battery Meter that depletes when driving/flying and recharges inside designated "Solar Microgrid Zones").

In doing so, your application must include:

- Smooth movement
- Directional movement
- Velocity
- Acceleration
- Trigonometric calculations ($\text{Math.sin}()$, $\text{Math.cos}()$)
- Battery or energy system
- Object-oriented programming

Use descriptive variable names.

Examples

✓ MissionScore

✗ y

2.2 OBSTACLES & COLLISION DETECTION (15 MARKS)

What you need to do

Create interactive environmental obstacles that represent real-world African infrastructure challenges. Your application must detect collisions mathematically and respond appropriately when the player interacts with obstacles or environmental hazards.

Include interactive obstacles such as:

- Potholes
- Rivers
- Wildlife
- Fallen trees
- Traffic
- Construction zones
- Load-shedding zones

Implement collision detection using:

- Rectangle (AABB), or
- Circular distance calculations

Environmental conditions should influence gameplay.

Examples:

- Rain reduces visibility.
- Wind changes movement.
- Load-shedding disables charging stations.

2.3 USER EXPERIENCE & ORIGINAL FEATURE (10 MARKS)

What you need to do: Design an engaging user experience using visuals, sound, and animation. You must also create at least one original feature without the assistance of Generative AI and explain its implementation in your project documentation.

Include:

- Background music or sound effects
- African-inspired visual theme
- Professional HUD
- Responsive interface

Original Feature

Develop one feature entirely yourself without using Generative AI.

Examples:

- Particle effects
- Dust animation
- Dynamic weather
- Day/night cycle
- Water ripple effects

Explain this feature in your README.

2.4 SCORE SYSTEM & LOCAL STORAGE (10 MARKS)

What you need to do

Implement a complete game management system that records player performance, saves scores locally, and allows users to restart or continue playing without refreshing the web page.

Your application must include:

- High scores
- Distance travelled
- Energy efficiency score
- Local storage
- Start screen
- Pause screen
- Game Over screen
- Restart without refreshing the browser

TASK 3: Presentation & Code Defence (20 Marks)

Brief Explanation

After completing your project, you will present your application to the lecturer. During the presentation, you must demonstrate that the application functions correctly, explain your programming decisions, and prove that you understand and own the code you submitted.

Each student will complete an individual 10-minute assessment.

3.1 LIVE DEMONSTRATION (6 MARKS)

Demonstrate:

- Running application
- Navigation
- Movement
- Physics
- Collision detection
- Game states
- Source code explanation

3.2 GITHUB & AI VERIFICATION (6 MARKS)

What you need to do

Present your completed application and demonstrate its main features. Explain how the application works, including movement, collision detection, game logic, and key programming concepts used during development.

The lecturer will review:

- GitHub commit history
- Repository structure
- AI Reflection Log

You must explain how AI-generated suggestions were modified to fit your own project.

3.3 LIVE CODING CHALLENGE (8 MARKS)

What you need to do

Complete a short coding exercise during your presentation. The lecturer will ask you to modify or extend part of your application to demonstrate your understanding of the submitted code and your ability to solve programming problems independently.

During the presentation you will complete a short coding exercise.

Examples:

- Increase obstacle speed.

- Modify collision behaviour.
- Change battery consumption.
- Add a keyboard shortcut.
- Adjust movement physics.

This activity demonstrates your understanding and ownership of the submitted code.

Monitoring Days are **compulsory progress checkpoints** that help ensure you are developing your project consistently throughout the semester. During each session, you must demonstrate your current progress and receive feedback before continuing to the next stage of development.

MONITORING DAYS (10 MARKS)

Monitor Day 1 (5 Marks)

Demonstrate: 26/08/2025

- GitHub repository
- Project structure
- HTML5 Canvas setup
- Basic player movement
- Approximately 50% of the core functionality

Monitor Day 2 (5 Marks)

Demonstrate: 16/09/2026

- Obstacles
- Collision detection
- HUD
- African-themed assets
- Additional gameplay mechanics

Submission Checklist

Submit the following:

- ❖ Public GitHub Repository URL
- ❖ GitHub Pages live application link
- ❖ README.md
- ❖ African Context Report
- ❖ Wireframe
- ❖ AI Reflection Log
- ❖ Fully functional HTML5 Canvas application

Academic Integrity

This is an individual assessment.

You may use Generative AI as a learning and support tool, but you must:

- Acknowledge all AI assistance in your AI Reflection Log.
- Understand and be able to explain every line of submitted code.
- Demonstrate code ownership during the live presentation and coding challenge.

Failure to explain or modify your own code during the presentation may result in marks being deducted under Task 3 (Code Defence).

Due Date: WAS262 Assignment 28/09/2026

Total: 100 marks (Planning: 15 + Implementation: 55 + Presentation: 20 + Monitor days: 10).

FORMAT

This assignment will be submitted online; therefore, you will need to submit a complete project file folder containing both the project design files, documentation and project screenshots, as outlined in the instruction document. Ensure you upload your project work to Canvas as a Zip file.

Ensure documentation work is submitted neatly and laid out on an A4 page(s) in a Word document using Calibri 10 font format using 1.5 line spacing.

- **Word document** Project Report named *StudName_WAS262_SF1.docx*.
- **Web Animation Project files** + Documentation to be uploaded to Canvas as a single Zip File with your *StudName_WebAnime_WAS262_SF1.docx*.

TECHNICAL REQUIREMENTS

To successfully complete this assignment, you will need to make use of the following tools:

- Microsoft Word
- Visual Studio Code
- Canvas

CHECKLIST

Once you have completed your assessment, use the checklist below to ensure that you have covered all elements.

	Task/ Requirement	YES	NO
1	Ensure that you submit all deliverables including project: - Source Files - Project Screen Shot - Documentation Did you upload all project files in a neatly structured and organized and named appropriately?		
2	File Naming: Follows the specified format with all assignment files		
3	Evidence of AI Use: If applicable, include screenshots of prompts and responses. <i>Ensure you include all your planning documents. If AI is used, include screenshots of your prompts and responses. This evidence will ensure that your assessment is valid and authentic. Without this evidence, your assessment will not be marked.</i>		
4	Technical Tools Used: Visual Studio Code & MS Word.		
5	Font and Spacing: Use Calibri, 10-point, 1.5 line spacing for documents.		
6	Did you include a list of references / in-text referencing?		

Marking Rubric Guide:

Section 1: Planning, GitHub & Africanisation (20 Marks)

CRITERIA	NEEDS IMPROVEMENT	GOOD	EXCELLENT
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1.1 GitHub Repository Setup & Commit History (5 Marks)	Repository missing, incomplete, or fewer than 8 commits. Poor project structure. (0–2 Marks)	Repository created with basic structure and several meaningful commits, but development history lacks consistency. (3–4 Marks)	Repository is well organised, contains all required files, and demonstrates continuous development with 8+ meaningful commits over multiple days. (5 Marks)
1.2 African Problem Investigation & Mathematics Brief (5 Marks)	Problem poorly explained or lacks African context. Minimal explanation of mathematical concepts. (0–2 Marks)	African problem identified with reasonable explanation and some discussion of movement or physics calculations. (3–4 Marks)	Real-world African problem clearly analysed with strong justification and detailed explanation of mathematical modelling (vectors, movement, physics, etc.). (5 Marks)
1.3 Wireframe, AI Reflection & Planning (10 Marks)	Missing or incomplete wireframe layout. Little evidence of UI planning. (0–2 Marks)	Wireframe completed with basic layout, structural hierarchy, and functional component planning. (3–4 Marks)	Professional wireframe with clear, comprehensive UI planning, detailed layout structure, and design flow. (5 Marks)
1.3.1 AI Usage & Design Reflection (5 Marks)	AI Used: Copied AI design suggestions without evaluation.	AI Used: Shows basic evaluation of AI-generated UI ideas or code.	AI Used: Demonstrates critical evaluation and meaningful, documented improvement of AI-generated suggestions.
	(0–2 Marks)	(3–4 Marks)	(5 Marks)
<i>(Optional: If No AI was used, full marks apply based on manual design decisions)</i>	No AI Used: Unable to justify design and layout choices.	No AI Used: Provides reasonable explanation for layout and design choices.	No AI Used: Clearly articulates design decisions, UX rationale, and why AI generation was bypassed.

Section 2: Game Development & Africanisation (50 Marks)

CRITERIA	NEEDS IMPROVEMENT	GOOD	EXCELLENT
2.1 Player Movement, Physics & Resource Management (15 Marks)	Movement is incomplete or unrealistic. Little or no mathematical implementation. (0–6 Marks)	Player movement works correctly with basic mathematical calculations and functional battery/resource system. (7–11 Marks)	Smooth movement using vectors, trigonometry, acceleration, drag, and resource management with well-structured code and descriptive variables. (12–15 Marks)
2.2 African Obstacles & Collision Detection (15 Marks)	Few obstacles or collision detection is inaccurate or missing. (0–6 Marks)	Appropriate obstacles implemented with functional collision detection and basic environmental effects. (7–11 Marks)	Multiple African-inspired obstacles with accurate collision detection, environmental modifiers, and realistic gameplay interactions. (12–15 Marks)

2.3 User Experience, Visual Design & Original Feature (10 Marks)	Limited visuals, poor interface, minimal originality, or missing custom feature. (0–4 Marks)	Good visual presentation with sound effects and one original feature showing reasonable creativity. (5–7 Marks)	Professional interface, polished animations, contextual audio, strong African theme, and an innovative feature developed independently. (8–10 Marks)
2.4 Score System, Local Storage & Game States (10 Marks)	Scoring or game states incomplete. Local storage missing or unreliable. (0–4 Marks)	Functional scoring system, local storage, and game state transitions with minor issues. (5–7 Marks)	Complete score management, persistent local storage, seamless state transitions, and smooth replay functionality without page refresh. (8–10 Marks)

Section 3: Presentation & Code Defence (20 Marks)

CRITERIA	NEEDS IMPROVEMENT	GOOD	EXCELLENT
3.1 Live Demonstration & Architecture Explanation (6 Marks)	Unable to demonstrate major functionality or explain code effectively. (0–2 Marks)	Demonstrates most application features and explains key programming concepts with some assistance. (3–4 Marks)	Confidently demonstrates all major features and clearly explains architecture, movement, physics, and collision logic. (5–6 Marks)
3.2.1 GitHub History & Version Control (3 Marks)	Limited GitHub history or commit logs; unable to explain the development process. (0–1 Marks)	Explains GitHub commit history and development process with reasonable understanding. (2 Marks)	Clearly demonstrates continuous development with meaningful commit history and clear milestone tracking. (3 Marks)
3.2.2 AI Usage & Code Integration (3 Marks)	AI Used: Blindly copied code without understanding or ability to explain.	AI Used: Explains AI prompts and basic integration into the code.	AI Used: Justifies why AI was used, how output was critically evaluated, debugged, and integrated.
	(0–1 Marks)	(2 Marks)	(3 Marks)
<i>(Optional: If NO AI was used, marks will be applied based on manual development reflection)</i>	No AI Used: Unable to explain manual problem-solving choices.	No AI Used: Explains manual development and tool choices reasonably.	No AI Used: Articulates clearly why AI was bypassed and demonstrates strong manual problem-solving strategies.
3.3 Live Coding Challenge (8 Marks)	Unable to modify or debug the application independently. (0–3 Marks)	Successfully completes part of the coding challenge with some guidance. (4–6 Marks)	Independently modifies, debugs, or extends the application correctly while explaining the implemented solution. (7–8 Marks)

Section 4: Monitoring & Project Progress (10 Marks)

CRITERIA	NEEDS IMPROVEMENT	GOOD	EXCELLENT
Monitor Day 1 (5 Marks)	Little or no progress demonstrated. Repository or Canvas project incomplete. (0–2 Marks)	Repository established with working Canvas setup and approximately 50% of the planned functionality demonstrated. (3–4 Marks)	Repository well organised with consistent commits, functional Canvas implementation, and more than 50% of the required functionality completed. (5 Marks)
Monitor Day 2 (5 Marks)	Minimal additional development since Monitor Day 1. Key features incomplete. (0–2 Marks)	Demonstrates obstacle implementation, collision detection, and significant project progress. (3–4 Marks)	Project is nearly complete with working gameplay mechanics, African-themed assets, polished interface, and evidence of final refinements. (5 Marks)

Additional Readings

Web design has become increasingly important when it comes to motion. An animated interface adds layers of complexity, aiding the user, offering feedback, delighting, and engaging them along the way.

Valuable Links:

- <https://www.kirupa.com/>
- [Homepage | GSAP](#)
- [p5.js](#)

Below are some of the latest online articles to read up on additional information regarding web animation.

1. [How To JS Animate \(w3schools.com\)](#)
2. [Animation for Beginners \(Where do I start?\) | Bloop Animation](#)
3. [Basic animations - Web APIs | MDN \(mozilla.org\)](#)
4. [Using the Web Animations API - Web APIs | MDN \(mozilla.org\)](#)
5. [Web animations with HTML, CSS, and JavaScript - LogRocket Blog](#)
6. [JavaScript animations](#)
7. [Web Animations \(w3.org\)](#)
8. [HTML5 - Canvas \(tutorialspoint.com\)](#)
9. [CSS and JavaScript Animation Tips with Examples | by TRAN SON HOANG | Level Up Coding \(gitconnected.com\)](#)
10. [JavaScript Animations \(w3docs.com\)](#)